Maxwell's Speed Law via MGFs

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Repository: github.com/HongruZhao/MaxwellMGF (private). **Version:** Git tag v1.0.2. The repository includes pinned dependencies, build instructions, and verification logs.

1. Ideal-gas model and physical conclusion

Consider a classical ideal gas at thermal equilibrium, viewed in its *bulk rest frame*. Let $V = (V_1, V_2, V_3)$ be the velocity of a randomly chosen molecule and $R = \|V\|$ its speed. Following Maxwell's Proposition IV, we assume two properties of its velocity density [1]:

1. **Isotropy:** there is no preferred direction. The joint velocity density depends only on $v_1^2 + v_2^2 + v_3^2$.
2. **Independent components:** the three Cartesian velocity components are mutually independent. By isotropy they have the same one-dimensional density f [2].

Together these assumptions give

$$F(v_1, v_2, v_3) = f(v_1)f(v_2)f(v_3) = \Phi(v_1^2 + v_2^2 + v_3^2). \tag{1}$$

We also assume $f > 0$, f integrable with $\int_{\mathbb{R}} f(x)\,dx = 1$, and sufficient smoothness: $g = \log f$ and a real extension H of the radial log-density $\log \Phi$ are differentiable. These are explicit model assumptions; independent components are not a consequence of choosing perpendicular axes.

Physical conclusion (conditional on this model and the three cited background inputs on page 3). The molecule's velocity density must be an isotropic Gaussian: for some component standard deviation $\sigma > 0$,

$$F(v_1, v_2, v_3) = \frac{1}{(2\pi\sigma^2)^{3/2}} \exp\left(-\frac{v_1^2 + v_2^2 + v_3^2}{2\sigma^2}\right). \tag{2}$$

Thus each signed velocity component has a centered normal distribution with variance σ^2. The molecule's nonnegative *speed* has the Maxwell density

$$p_R(r) = \begin{cases} \sqrt{2/\pi}\, r^2 \sigma^{-3} e^{-r^2/(2\sigma^2)}, & r \geq 0, \\ 0, & r < 0, \end{cases} \qquad \int_{\mathbb{R}} p_R(r)\,dr = 1. \tag{3}$$

The assumptions determine these distributional shapes, while leaving the scale σ unspecified. Lean checks the actual velocity and speed laws and their normalization. Section 3 explains the MGF and chi-square steps connecting the two laws. Equilibration from collisions is outside the claim.

2. Deriving the Gaussian density

Taking logarithms of (1) gives $g(x) + g(y) + g(z) = H(x^2 + y^2 + z^2)$. Differentiating in x yields $g'(x) = 2xH'(x^2 + y^2 + z^2)$. Comparing $(x, 1, 0)$ with $(1, x, 0)$ gives $g'(x) = xg'(1)$, including $x = 0$. Hence $g(x) = g(0) - bx^2$, with $b = -g'(1)/2$. Integrability forces $b > 0$: otherwise $f(x) \geq e^{g(0)} > 0$ on all of $\mathbb{R}$. Normalization gives $f(x) = \sqrt{b/\pi}\, e^{-bx^2}$ and $\sigma^2 = (2b)^{-1}$, yielding (2).

3. From the joint MGF to chi-square laws

Define $U_i = (V_i/\sigma)^2$ and $S = \sum_{i=1}^3 U_i = R^2/\sigma^2$, the dimensionless squared speed. For every $t = (t_1, t_2, t_3) \in \mathbb{R}^3$, the Gaussian exponential integrals are finite. Independence therefore gives

$$M_V(t) = \mathbb{E}e^{\sum_i t_i V_i} = \prod_i e^{\sigma^2 t_i^2/2} = e^{\sigma^2 \|t\|^2/2}.$$

Measurable functions applied separately to mutually independent variables remain independent. Applying $x \mapsto (x/\sigma)^2$ thus preserves independence of all three coordinates. With $Z_i = V_i/\sigma$, the one-dimensional Gaussian integral yields, for $t < 1/2$,

$$M_{U_i}(t) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{-(1-2t)z^2/2}\, dz = (1-2t)^{-1/2}. \tag{4}$$

The coefficient $1 - 2t > 0$ also proves absolute integrability. The Lean development evaluates the same integral in the equivalent parameterization $U_i = 2bV_i^2$; it does not assume the square-normal law.

Throughout, Gamma(a, r) uses *shape a and rate r*, with density

$$q_{a,r}(s) = \frac{r^a}{\Gamma(a)} s^{a-1} e^{-rs}\, \mathbf{1}_{\{s>0\}}.$$

Its MGF is $(1 - t/r)^{-a}$ for $t < r$ [4]. By (4), MGF uniqueness identifies U_i with Gamma$(1/2, 1/2) = \chi^2_1$. Multiplication of the three MGFs gives

$$M_S(t) = \prod_{i=1}^3 M_{U_i}(t) = (1-2t)^{-3/2}, \qquad t < 1/2. \tag{5}$$

The code also proves exponential integrability of the sum. Both comparison MGFs are finite on $|t| < 1/4$. Uniqueness therefore gives

$$\mathcal{L}(S) = \text{Gamma}(3/2, 1/2) = \chi^2_3.$$

This identifies the sum as a chi-square law [3, 5]. Equality of moments without this existence argument would not suffice.

4. Transforming squared speed into speed

Since $S = R^2/\sigma^2$ and $R \geq 0$, $R = \sigma\sqrt{S}$ pointwise. For $r > 0$, change variables $s = r^2/\sigma^2$, with $ds/dr = 2r/\sigma^2$. Using $\Gamma(3/2) = \sqrt{\pi}/2$ gives

$$p_R(r) = q_{3/2,\,1/2}(r^2/\sigma^2)\frac{2r}{\sigma^2} = \sqrt{2/\pi}\,\frac{r^2}{\sigma^3} e^{-r^2/(2\sigma^2)}.$$

There is no atom at zero, and the density is zero for negative speeds. Normalization follows from the same substitution: $\int_0^\infty p_R(r)\, dr = \int_0^\infty q_{3/2,\,1/2}(s)\, ds = 1$.

The formal proof checks the density independently by polar integration of the derived Gaussian velocity density: $\int_{\mathbb{R}^3} F(\|v\|)\, dv = 4\pi \int_0^\infty r^2 F(r)\, dr$. It proves both $\mathcal{L}(R) = p_R(r)\, dr$ and the explicit transformation identity

$$(s \mapsto \sigma\sqrt{s})_* \,\text{Gamma}(3/2, 1/2) = p_R(r)\, dr.$$

The latter follows by composing the checked pushforward for S with the pointwise identity for R. Thus the distribution statements include the actual laws and their normalization.

5. External inputs, verification, and scope

Exactly **three additional mathematical axioms** are declared; all are generic background results, and none states the Maxwell characterization, the three-square sum law, or the speed-law conclusion.

A1. Product-density criterion. For any finite dimension n, a measurable $W : \Omega \to \mathbb{R}^n$ with density $\prod_i f(w_i)$, where f is measurable, nonnegative, integrable, and $\int f = 1$, has mutually independent coordinates with common law $f(x)\,dx$. Used to pass from the physical density model to the independence API and marginal measures [2, independence between continuous variables; mutual independence].

A2. Gamma MGF. For every $a, r > 0$ and $t < r$, e^{tx} is integrable under Gamma(a, r) and its integral is $(1 - t/r)^{-a}$. Used to identify the comparison MGFs at $(a, r) = (1/2, 1/2)$ and $(3/2, 1/2)$ [4, general Gamma distribution, Moments].

A3. MGF uniqueness. If measurable X on a probability space and a probability measure ν have finite equal MGFs for every $|t| < \varepsilon$, for some $\varepsilon > 0$, then $\mathcal{L}(X) = \nu$. Used for the one-square and three-square laws, with $\varepsilon = 1/4$ [3, characterization of a distribution via the MGF].

The physical hypotheses in (1) and the assumed velocity density are theorem parameters. They are separate from A1–A3 and remain assumptions even when the conditional theorem is checked. Gaussian integration, differentiation, polar integration, Gamma normalization, preservation of independence by measurable maps, and independent-MGF multiplication use proved mathlib results [6].

Verification. Lean v4.33.0-rc2 and mathlib revision `641fbd329d4ffb62bef83c51f54088469056bd36` are pinned. The project builds without errors or proof holes. Its axiom audit exposes A1–A3 in the complete theorem, together with the standard foundations `propext`, `Classical.choice`, and `Quot.sound`. The Gaussian characterization and direct speed pushforward need only those standard foundations. Complete theorem statements, source-to-report correspondence, and build/audit logs accompany the source.

Limitations and delivery status. This proves a smooth positive-density characterization conditional on the model and A1–A3. It does not derive independence or equilibration from collision mechanics, prove a minimal-regularity result, or calibrate σ to temperature. The repository is private; the instructor must have access to open and download the specified version. Team eligibility is an administrative requirement, not a mathematical conclusion.

AI and reused work. OpenAI Codex assisted with mathematical exposition, Lean code, debugging, and LaTeX preparation. The existing local Maxwell demonstration supplied the credited calculus and polar-integration core. The author must review the assumptions and be able to explain the argument; compiler acceptance does not validate the physical postulates.

References

[1] J. C. Maxwell (1860), *Illustrations of the Dynamical Theory of Gases*, Part I, Proposition IV. Supplied reprint, pp. 152–154 (PDF pp. 5–7). Source scan.

[2] M. Taboga, *Independent random variables*, StatLect.

[3] M. Taboga, *Moment generating function*, StatLect.

[4] K. Siegrist, *The Gamma Distribution*, Random: Probability, Mathematical Statistics, Stochastic Processes. The source uses scale b; here rate $r = 1/b$.

[5] K. Siegrist, *The Chi-Square Distribution*, same text, distribution and chi-distribution sections.

[6] The mathlib community, *mathlib4*, pinned revision above. Exact library declarations are recorded in the project documentation.